

PCB Designing Using Altium Designer

Mini Project - I

Submitted in partial fulfillment of the academic requirement for the award of
the degree of

BACHELOR OF ENGINEERING
In
ELECTRONICS AND COMMUNICATION
ENGINEERING

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2024-2025



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CERTIFICATE

This is to certify that the Mini project – I titled “PCB Designing Using Altium Designer” submitted by

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students of Electronics and Communication Engineering Department, Vasavi College of Engineering in partial fulfillment of the requirement for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering is a record of the bonafide work carried out by them during the academic year 2024-2025. The result embodied in this project report has not been submitted to any other university or institute for the award of any degree.

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This is to state that the work presented in this thesis titled **“PCB DESIGNING USING ALTIUM”** is a record of work done by us in the Department of Electronics and Communication Engineering, Vasavi College of Engineering, Hyderabad. No part of the thesis is copied from books/journals/internet and wherever the portion is taken, the same has been duly referred to in the text. The report is based on the project work done entirely by us and not copied from any other source. I hereby declare that the matter embedded in this thesis has not been submitted by me in full or partial thereof for the award of any degree/diploma of any other institution or university previously.

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ABSTRACT

This Project aims to design a Printed Circuit Board (PCB) in software named “Altium Designer”. The main objective of this mini project is to create an efficient Printed Circuit Board Layout to meet the industry standards. This involves both Single Layer and Multi-Layer. The circuit used for implementing Single Layer is LED CHASER CIRCUIT. The circuit used for implementing multi-layer is the SMART HOME AUTOMATION SYSTEM. Altium Designer is a professional Printed Circuit Board Design tool that supports advanced features like Schematic Design, Component Library, Footprints, Auto and Manual Routing, 3D visualization and generation of Gerber files.

In this project, The Schematic diagram was placed in the workspace using component library, even though all the components weren't available in Altium Designer component library, we used few external sources to place all the components according to the schematic diagram.

After the Schematic is placed, footprints have been created accordingly and for the simulation of single layer printed circuit board design LED chaser circuit, we used Proteus software to check for any errors and working of the designed schematic.

As Altium designer offers both auto and manual routing, the single latter has been done with the help of auto routing, while the multi-layer has been done with manual routing to maintain signal integrity. The final PCB layout was checked by Designed Rule check (DRC) and Gerber files were generated for Printed Circuit Board Fabrication.

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CHAPTER – 1

INTRODUCTION

Home automation, also known as smart home technology, involves the automatic control of 3 electronic devices within a household. As technology advances, there has been a significant shift toward connected homes where appliances, lights, heating, cooling, and security systems can be controlled remotely. This transformation has been largely driven by the rise of Internet of Things (IoT) devices, which allow seamless communication between hardware and software components over the internet.

1.1 Aim

To design and implement a home automation PCB circuit using ESP32 and Altium Designer.

1.2 Objectives

- To design a reliable ESP32-based PCB in Altium Designer.
- To control appliances using relay modules.
- To interface switches for manual control.
- To integrate proper power supply circuitry.

1.3 Methodology

- Schematic design in Altium.
- Component selection and simulation.
- PCB layout and routing.
- Fabrication and testing.

CHAPTER -2

LITERATURE REVIEW

Printed Circuit Boards acts as a backbone in the modern day electronic systems. To design a PCB in software in the growing technology, Altium Designer plays as one of the most important software tool. Several tools highlight Altium Designer's role in modern technology which includes Real – Time DRC (Design Rule Check) , Component Library management, integrated 3d modelling and many more additional features. The single layer (LED CHASER CIRCUIT) , explains to optimize trace width and component placement on single layer and acts as best practice for a beginner before we get into the main multi layer design.

In multi layer, we used Smart Home Control System circuit with the help of ESP32 micro controller with built-in WiFi or bluetooth for IoT- based smart home applications. These help in real time scenarios like appliance control via mobile apps / sensors, real time data transmission over Wi-Fi. So, this project highlights about role of Altium Designer in developing both single layer and multi layer Printed Circuit Board's by providing a strong foundation in embedded system design and pcb manufacturing design.

Chapter 3

Block Diagram/Flow Diagram

3.1 Block Diagram

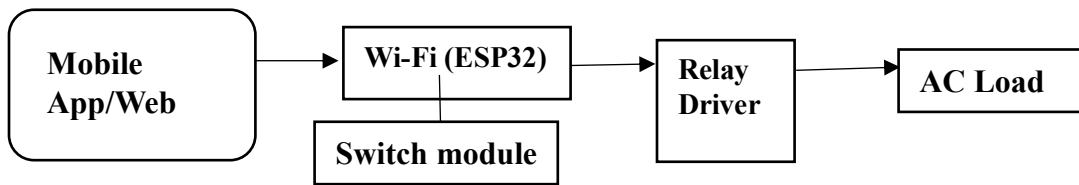


Figure 3.1 Block Diagram

3.2 Explanation

- ESP32 acts as the central controller.
- Relay modules switch AC appliances on/off.
- Manual switches provide physical control.
- The power supply provides 3.3V and 5V regulated outputs.

Hardware Components

- Microcontroller: ESP32-WROOM-32
- Modules: Relay Modules
- Power Supply: HLK-PM03 – AC to 3.3V DC module, Bridge Rectifier
- Passive Components: Resistors, Capacitors

Chapter 4

Project Implementation

Single Layered PCB

Hardware Components

- 555 Timer IC
- Decade Counter
- LED's (LED 1- LED 10)
- Passive components: Resistors,Capacitors
- 2-Pin connector.

Schematic Design

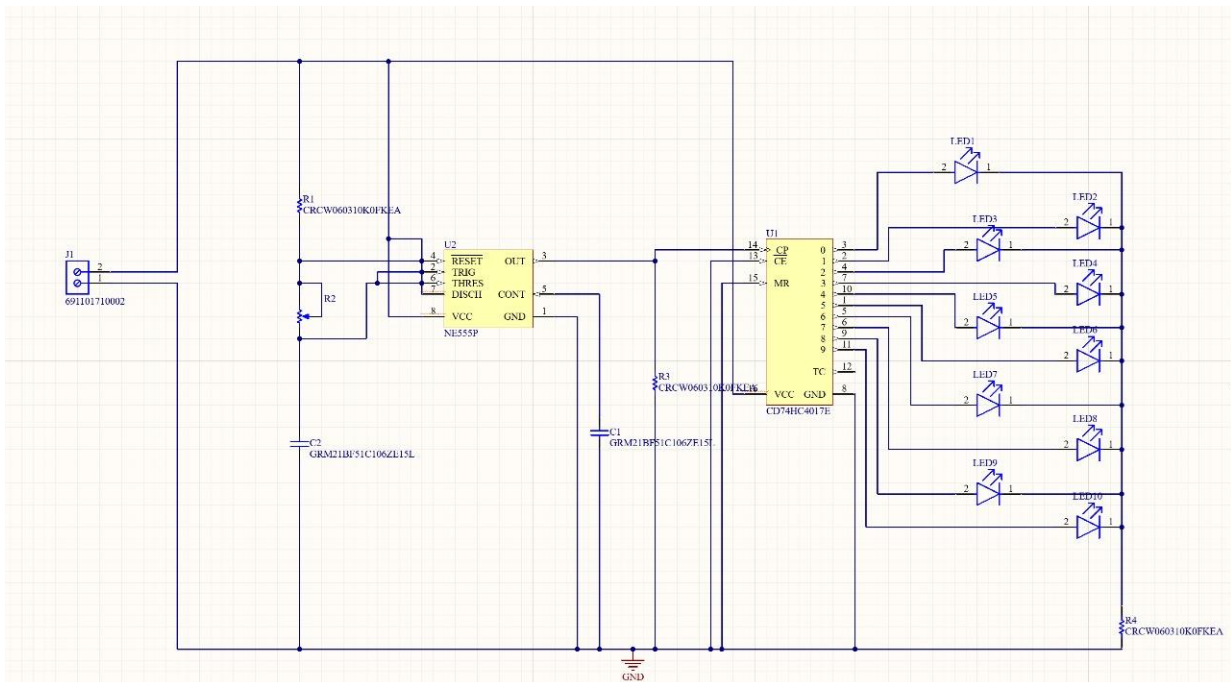


Figure 4.1 LED Chaser Circuit

The LED Chaser circuit uses a 555 Timer in astable mode to generate continuous clock pulses and these clock pulses are sent to the decade counter consisting of 10 LED's. Every clock pulse turns output on, one at a time and the LED's gets blinked one at a time sequentially. As one LED blinks only after its previous gets blinked, this is known as chasing and that's why we terminate the circuit as LED Chaser circuit.

When the last LED lights up, the counter gets reset and the sequence gets repeated eventually. Each LED has a current-limiting resistor to protect it from over current. The power is given through the 2 pin connector, supplying vcc and giving gnd to the whole circuit.

After placing the components and creating footprints, we preferred auto routing over manual routing because the circuit is simple and without any errors, Altium Designer has an advantage to automatically route without any involvement.

When the crucial part routing was done, we checked for any errors using DRC (Design rule check) and for the simulation, we installed Proteus software to check for the working and efficiency of our design. The LED's successfully got lightened up, one after the other making it Chase throughout the process and counter got reset and the sequence had repeated continuously.

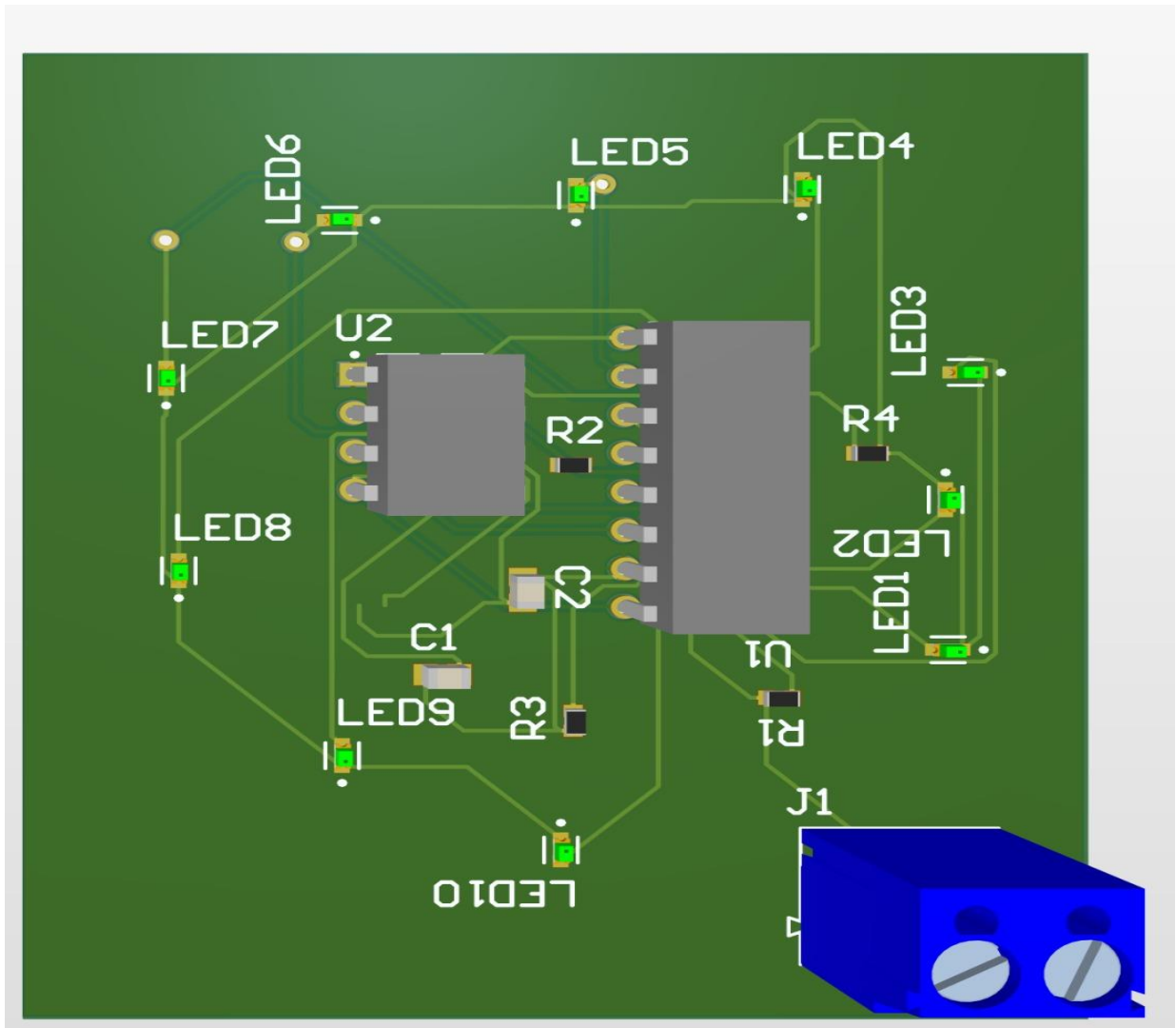


Figure 4.2 3D view of single layered PCB

The above two figures represent 3D visualisation and Gerber files respectively of the single layer LED Chaser circuit. Altium Designer is a powerful tool which allows us to use advanced features which include footprints, supports both auto and manual routing, 3D generation of the circuit and finally the gerber files generation.

After successfully implementing the circuit, we checked for any errors in the schematic. We checked from Import From Layout option to know if any of the errors exist or not. Later, we created footprints for the schematic. As we move on , routing is the next thing to be done which is most crucial part about designing the PCB.

As said earlier, Altium Designer supports both auto and manual routing. For this single layer LED chaser circuit, we went for auto routing because it's not a complex circuit to adjust any changes. The circuit was simple and clear.

Now, let's dive into the main part of the PCB design, Multi Layer PCB Design. The circuit which we've selected is Home Automation system control using ESP32 micro-controller. Here is a detail list and working of the multi layer:

Schematic Design

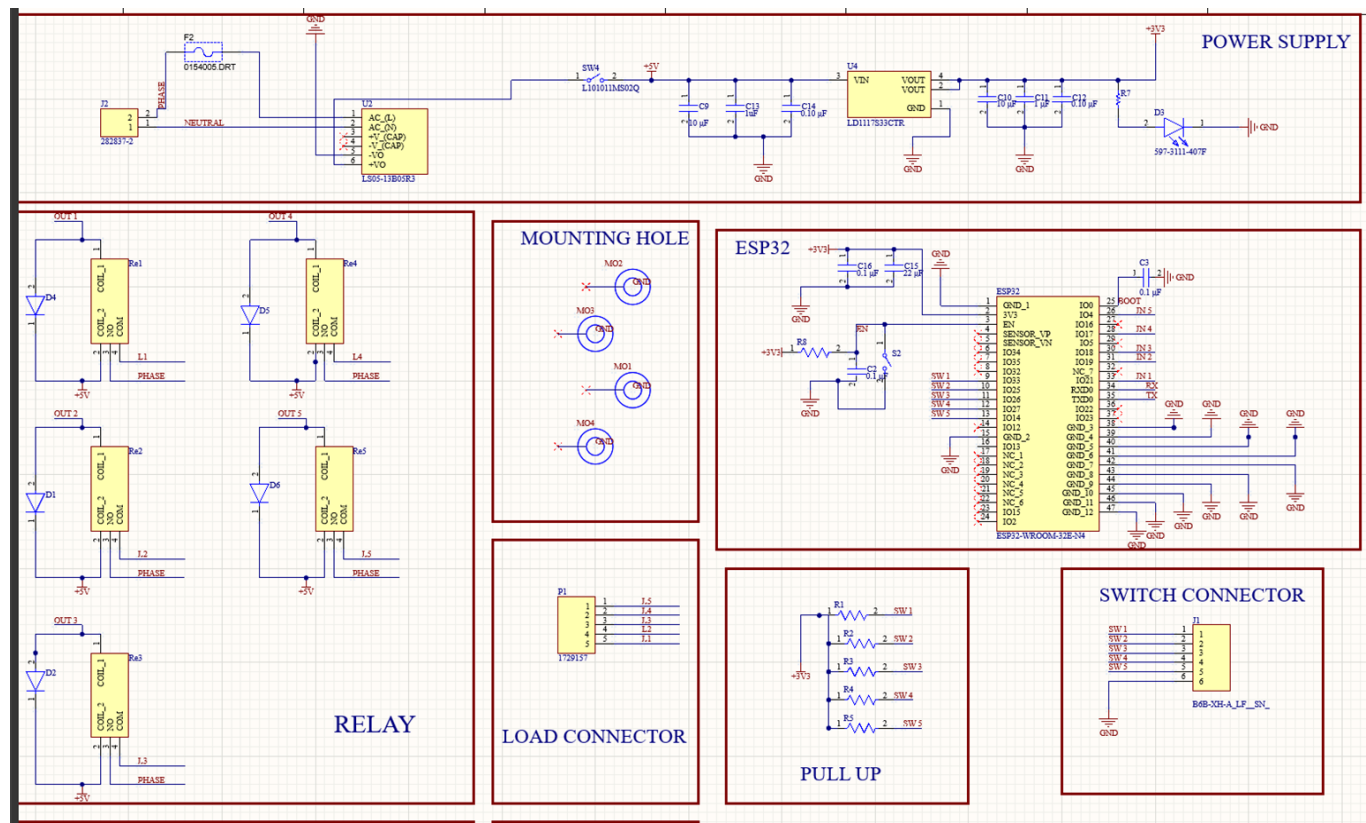


Figure 4.3 Smart Home Automation

Six Layer PCB Stack-Up

#	Name	Material	Type	Weight	Thickness	Dk	Df
	Top Overlay		Overlay				
	Top Solder	Solder Resist	Solder Mask		0.4mil	3.5	
1	Top Layer		Signal	1oz	1.4mil		
	Dielectric 4	PP-006	Prepreg		2.8mil	4.1	0.02
2	GND	CF-004	Signal	1oz	1.378mil		
	Dielectric 2	PP-006	Prepreg		2.8mil	4.1	0.02
3	5V	CF-004	Plane	1oz	1.378mil		
	Dielectric 1	FR-4	Dielectric		12.6mil	4.8	
4	3.3V	CF-004	Plane	1oz	1.378mil		
	Dielectric 3	PP-006	Prepreg		2.8mil	4.1	0.02
5	Layer 4	CF-004	Signal	1oz	1.378mil		
	Dielectric 5	PP-006	Prepreg		2.8mil	4.1	0.02
6	Bottom Layer		Signal	1oz	1.4mil		
	Bottom Solder	Solder Resist	Solder Mask		0.4mil	3.5	
	Bottom Overlay		Overlay				

Figure 4.4 Layer Stack Manager

Layer	Purpose
Top Layer	Components + Signal
Inner Layer 1	Signal + Ground
Inner Layer 2	Power Plane (5V)
Inner Layer 3	Power Plane (3.3V)
Inner Layer 4	Signal + Routing
Bottom Layer	Components + Signal

The advantage of having these multilayer stack-up is that it isolates the 5V and 3.3V power planes effectively diminishes noise and minimizes ripple in the power supply. By providing lower impedance pathways for power delivery, these planes enhance the efficiency of the power distribution network. Additionally, the planes function like distributed capacitors—an asset for suppressing high-frequency noise, particularly advantageous for fast-switching integrated circuits (ICs).

It provides more routing space for complex designs and also controlled impedance for high-speed and good thermal management and symmetrical design for manufacturing

PCB Layout and Routing

Top Layer

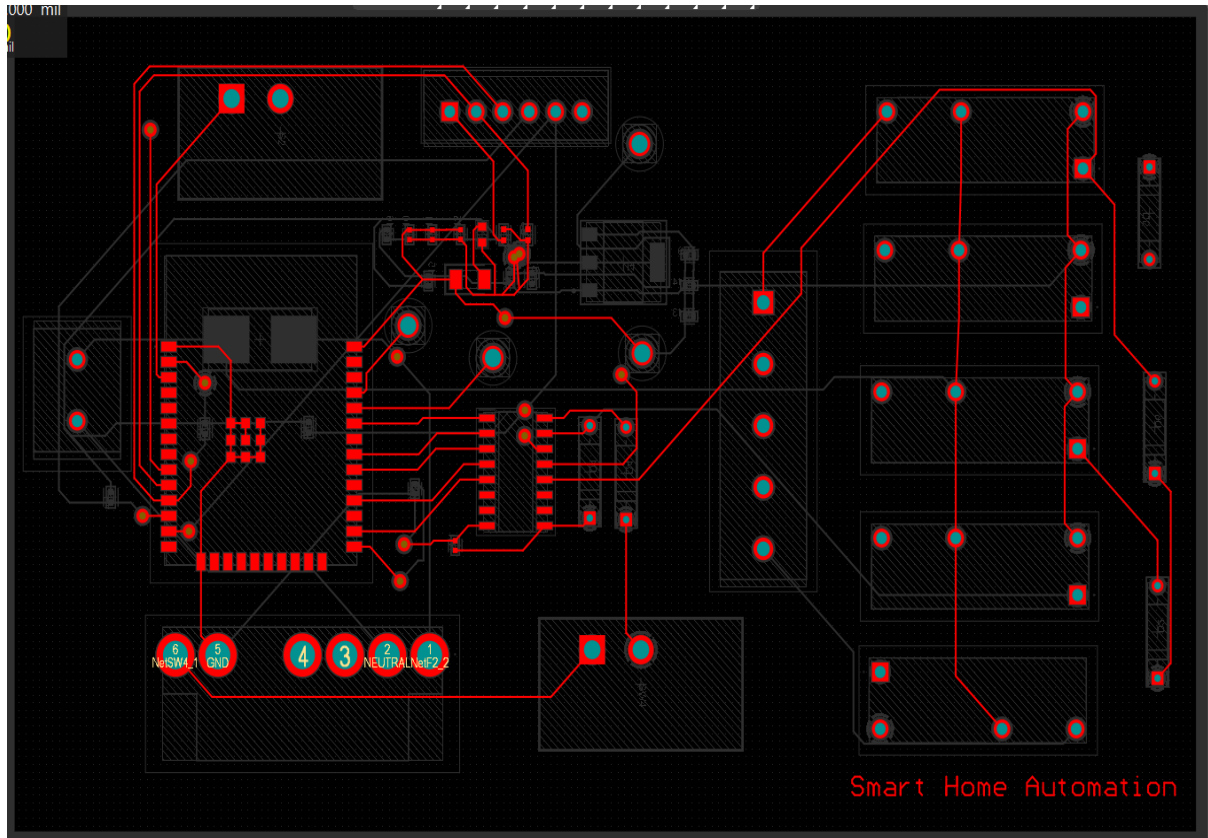


Figure 4.5 Top Layer

The top layer has the primary functional components, including the ESP32 module, relays, and key connectors. Placing the ESP32 on top ensures optimal Wi-Fi signal strength and easy access for programming and debugging. The relays are also on this layer to allow efficient heat dissipation, physical separation for AC loads, and straightforward routing from the microcontroller. Connectors for switches and load outputs are positioned here for convenient user interfacing and clean signal routing. This toplayer layout prioritizes accessibility, RF performance, and logical signal.

Bottom Layer

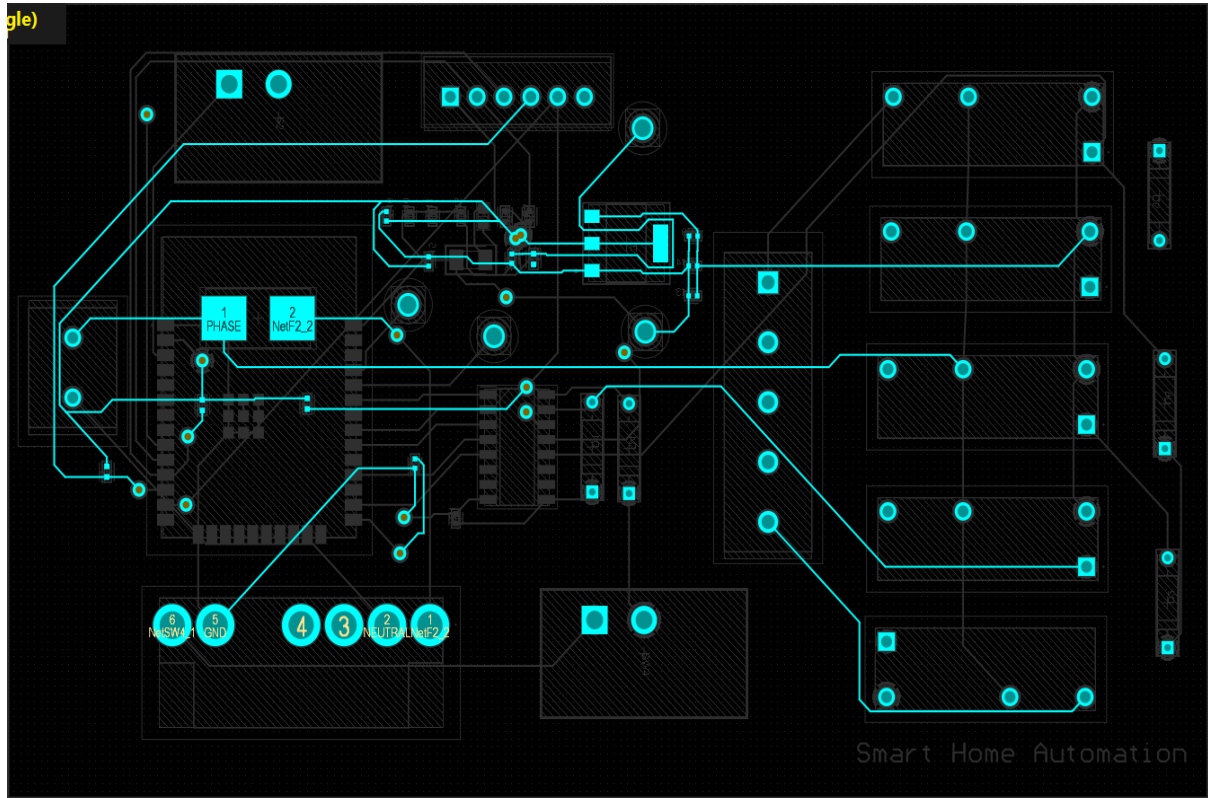


Figure 4.6 Bottom Layer

The bottom layer is dedicated to supporting things like flyback diodes for the relays, the voltage regulator and filtering capacitors, pull-up resistors for switches, and terminal blocks for external connections. These components are placed here to reduce congestion on the top layer, improve thermal distribution, and simplify routing. By locating less interactive components and passive elements on the bottom, the design maintains a clean top-side layout while ensuring functional separation and better electrical performance.

3D View of Top and Bottom layer of the PCB

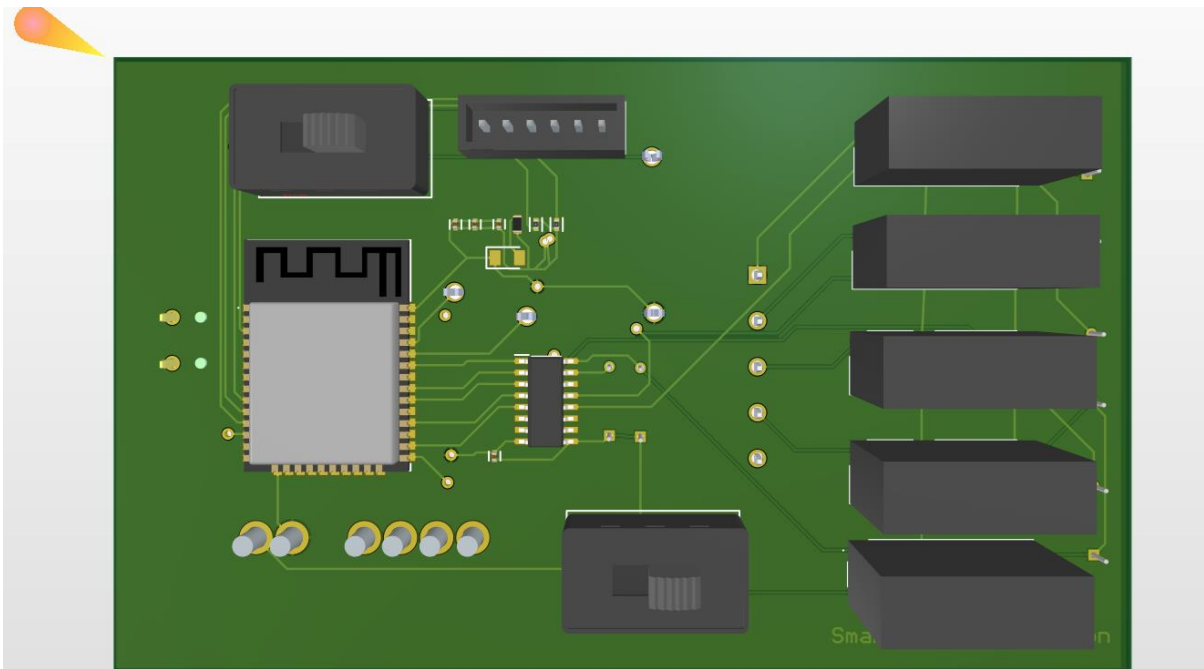


Figure 4.7 Top Layer

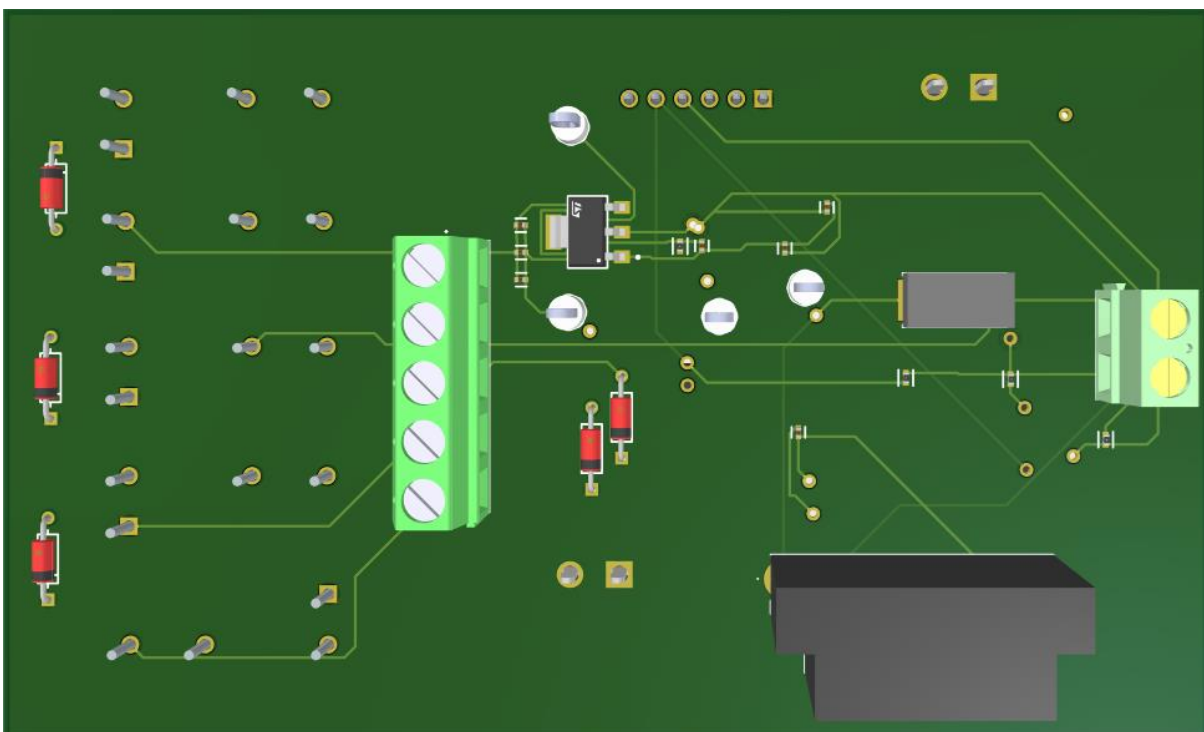


Figure 4.8 Bottom Layer

Bill of Materials

1	Comment	Description	Designator	Footprint	Quantity
2	C0402C104K8PACTU	CAP CER 0.1UF 10V X5R	C1, C2, C3, C9, C10, C11	FP-C0402-BB-MFG	11
3	1N4148	DIODE GEN PURP 100V	D1, D2, D4, D5, D6	FP-017AG-MFG	5
4	597-3111-407F	LED	D3	LEDC3015X150N	1
5	ESP32-WROOM-32E-N	Integrated Circuit	ESP32	ESP32WROOM32EN4	1
6	0154005.DRT	Fuse and Holder Comb	F2	LFUS-154_V	1
7	B6B-XH-A_LF__SN_	Connector	J1	SHDR6W64P0X250_1X	1
8	282837-2	Connector	J2	2828372	1
9	1729157	Connector	P1	1729157	1
10	CRCW040210K0JNED	RES Thick Film, 10kΩ, 5	R1, R2, R3, R4, R5, R8	FP-CRCW0402-e3-IPC	6
11	ERJ-3EKF3300V	Chip Resistor, 330 Ohm	R7	RESC1608X55X30LL15	1
12	HF46F-G_5-HS1T	Relay or Contactor	Re1, Re2, Re3, Re4, Re5	HF46FG5HS1T	5
13	L101011MS02Q	SWITCH SLIDE SPST 4A	S2, SW4	FP-L101011MS02Q-MF	2
14	TEST	Test Point, White, Throu	TP1, TP2, TP4, TP5	KSTN-5012_V	4
15	LS05-13B05R3	Power Supply	U2	LS0513B05R3	1
16	LD1117S33CTR	IC REG LINEAR 3.3V 800	U4	FP-SOT-223-0046067_	1
17	ULN2003LVDR	Integrated Circuit	U5	SOIC127P600X175-16N	1

Figure 4.9 Bill of Materials

Chapter 5

Results

5.1 Functional Testing (Software-Level Verification)

The schematic and PCB layout were thoroughly verified in Altium Designer. ERC (Electrical Rule Check) and DRC (Design Rule Check) confirmed correct connections, clearances, and design compliance. Netlist comparison ensured consistency between schematic and PCB layout. SPICE simulations validated the power supply and relay circuits. A 3D clearance check rule confirmed proper component placement with no physical interferences. The BOM was reviewed for availability and accuracy. All tests were passed successfully, and the design was approved for fabrication.

Gerber Files

The yellow layer indicates the copper layer of polygon pour plot for connecting to ground.

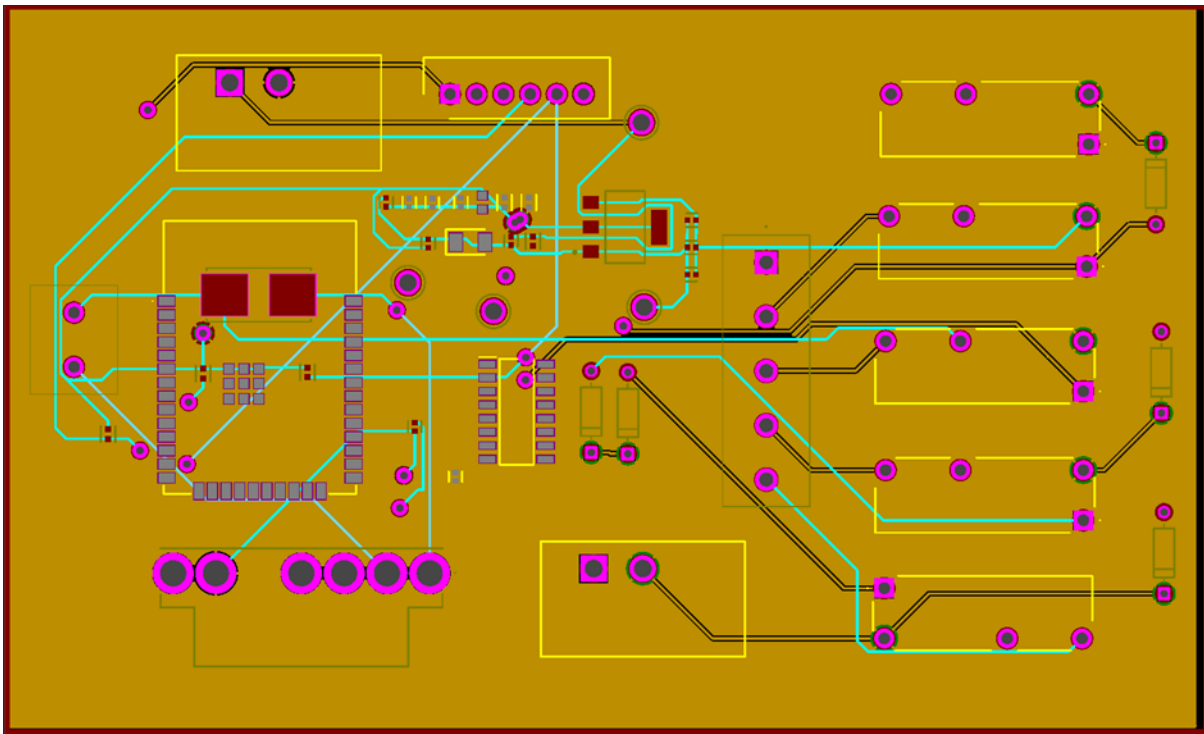


Figure 5.1 Multi-Layer PCB

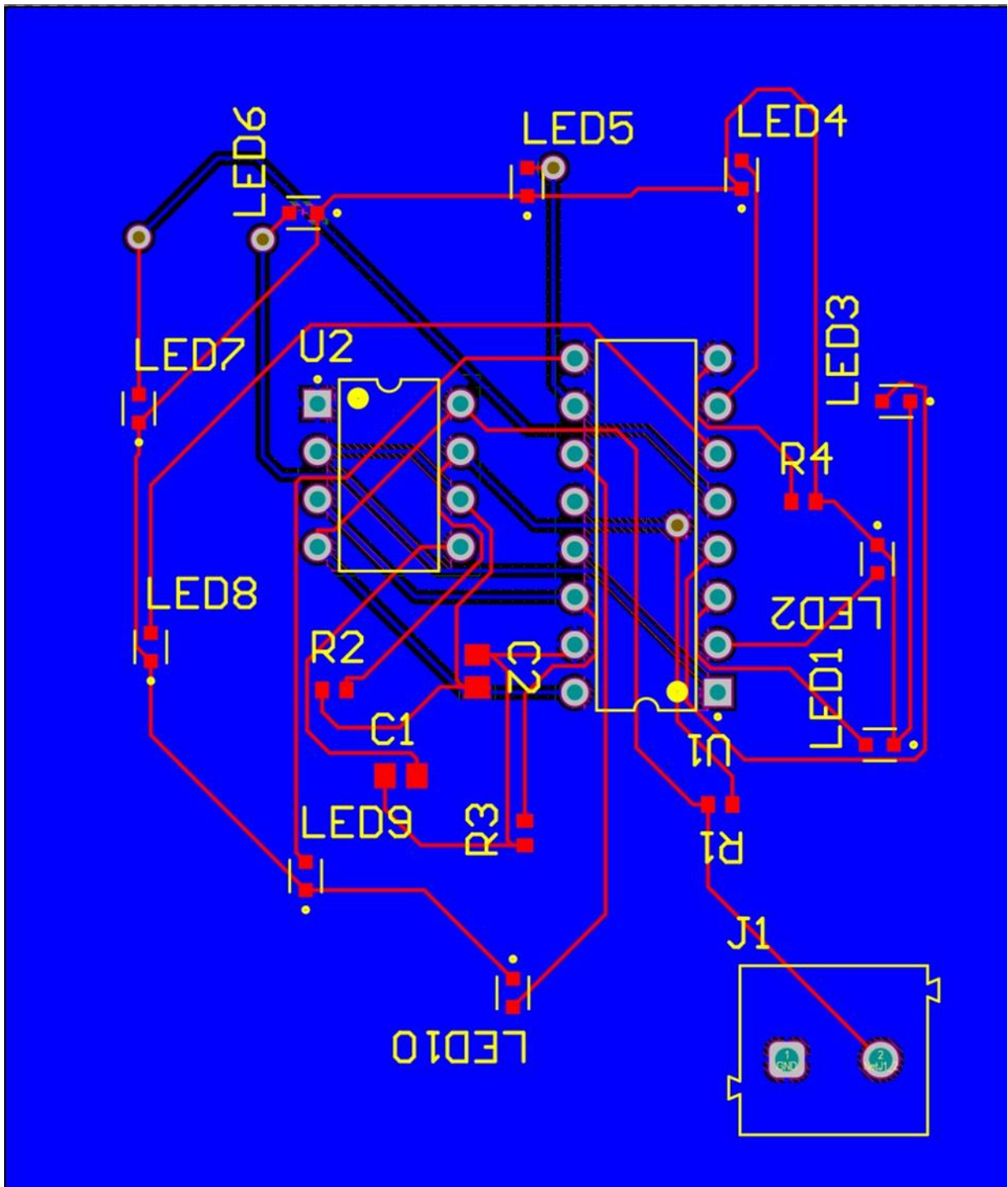


Figure 5.2 Single Layer PCB

Conclusion

The PCB Design for the LED Chaser Circuit (Single Layer) and Smart Home Automation system (multi-layer) have been implemented successfully in Altium Designer. This project demonstrates about the use of Altium Designer where Schematic Diagram implementation using components library, creating footprints, auto and manual routing, Design Rule Check can be used.

By using ESP32 Micro-controller, with various automation modules such as relays, sensors, mounting holes, connecting pins, diodes, phase, load connectors and switch connectors, a compact and reliable hardware layout can be achieved by the generation of gerber files.

The final design ensured efficient power distribution and maintained proper spacing as per the design rules. The use of 3D Visualization in the Altium Designer helps rule-checking features increases the accuracy and manufacturability of the board. This project not only indicates about the fundamentals of PCB design, but also showcases about modern tools usage in the rapid development of IoT-based automation systems.

In conclusion, ESP-32 Micro Controller sends signals to the relays based on commands and instructions from Mobile app or any sensors and it acts as brain of the automation system. So basically, the ESP32 connects to a home Wi-Fi network and a user sends instruction/command from a mobile app or sensor. Here, The ESP32 Micro-controller processes the command and it activates the relays. The relay switches ON/OFF the appliances like lights, fans, AC's, etc. So, finally the user gets real-time feedback on the board and this is how the Printed Circuit Board using Smart Home control system circuit can be used in real-time applications.

Future Scope

- Add support for Bluetooth/MQTT.
- Integrate sensors (temperature, motion).
- Develop mobile app with better UI.
- Cloud-based control and analytics.
- Local Control via Physical Switches.

REFERENCES

1. Espressif Systems, *ESP32 Datasheet*, [Online]. Available: <https://www.espressif.com/en/support/download/documents>
2. Altium LLC, *Altium Designer Documentation*, [Online]. Available: <https://www.altium.com/documentation>
3. IEEE, *Papers on Smart Home Systems*, IEEE Xplore Digital Library, [Online]. Available: <https://ieeexplore.ieee.org/Xplore/home.jsp>
4. Stack Overflow, Reddit Embedded Systems Community, *Technical Discussions and Troubleshooting Resources*, [Online]. Available: <https://stackoverflow.com/> and <https://www.reddit.com/r/embedded/>
5. EEVblog Forum, *Altium Designer Tips and Discussions*, [Online]. Available: <https://www.eevblog.com/forum/altium/>
6. Simon Monk, *Make Your Own PCBs with Eagle: From Schematic Designs to Finished Boards*, McGraw-Hill Education, 2014.
7. Rui Santos, *Home Automation Projects with ESP32 and ESP8266*, Random Nerd Tutorials, [Online]. Available: <https://randomnerdtutorials.com/>
8. Andreas Spiess (YouTube Channel), *ESP32 and IoT Tutorials for Smart Home Projects*, [Online]. Available: <https://www.youtube.com/c/AndreasSpiess>
9. Espressif Systems, *ESP-IDF Programming Guide*, [Online]. Available: <https://docs.espressif.com/projects/esp-idf/en/latest/esp32/>
10. Neil Kolban, *Kolban's Book on ESP32*, [Online]. Available: <https://leanpub.com/kolban-ESP32>